YANGON UNIVERSITY OF ECONOMICS

DEPARTMENT OF STATISTICS

MASTER OF APPLIED STATISTICS PROGRAMME

ANALYZING THE RELATIONSHIP BETWEEN 24 CARAT GOLD PRICE AND CONSUMER PRICE INDEX IN MYANMAR (2012-2022)

KAUNG SI THU

MAS -1

MAS 5th BATCH

JUNE, 2025

YANGON UNIVERSITY OF ECONOMICS
DEPARTMENT OF STATISTICS
MASTER OF APPLIED STATISTICS PROGRAMME

ANALYZING THE RELATIONSHIP BETWEEN 24 CARAT GOLD PRICE AND CONSUMER PRICE INDEX IN MYANMAR (2012-2022)

Supervised By:

Dr. Win Naing Oo
Associate Professor
Department of Applied Statistics
Yangon University of Economics

Submitted By:

Kaung Si Thu
MAS – 1
MAS 5th Batch
Yangon University of Economics

JUNE, 2025

ACCEPTANCE

Accepted by the Board of Examiners of the Department of Statistics, Yangon University of Economics in partial fulfillment for the requirement of the Master Degree, Master of Applied Statistics.

.....…….
(Chairperson)
Prof. Dr Tin Tin Htwe
Rector
Yangon University of Economics

……
(Examiner)
Prof. Dr. Kyaw Tun Naing
Pro-Rector
Yangon University of Economics

…….
(Examiner)
Daw Khin Nu Win
Associate Professor (Retd.)
Department of Statistics
Yangon University of Economics

……
(Examiner)
Dr. Aye Thida
Professor/ Head
Department of Statistics
Yangon University of Economics

…….
(Examiner)
Dr. Hlaing Hlaing Moe
Professor/Head
Department of Applied Statistics
Yangon University of Economics

……
(Examiner)
Dr. Khin Thet Tun
Associate Professor
Department of Statistics
Yangon University of Economics

…….
(Supervisor)
Dr. Win Naing Oo
Associate Professor
Department of Applied Statistics
Yangon University of Economics

JUNE, 2025

ABSTRACT

This study examines the dynamic relationship between 24-carat gold prices and the Consumer Price Index (CPI) in Myanmar from April 2012 to July 2022, a period marked by economic instability, currency depreciation, and political upheaval. Gold is traditionally viewed as a hedge against inflation, but its effectiveness in Myanmar's unique economic context remains underexplored. Using monthly data sourced from Myanmar's Central Statistical Organization (CSO), the study employs advanced time series econometric techniques, including unit root tests, cointegration analysis, Vector Autoregressive Model (VAR), and Granger causality tests. Descriptive statistics reveal stark differences in volatility and show that gold prices moved up and down sharply, while CPI increased at a more stable pace. Unit root tests confirm that both variables are non-stationary at level but achieve stationarity after first differencing. Granger causality analysis further identifies a unidirectional relationship, with gold prices influencing future CPI movements, suggesting that gold serves as a leading indicator of inflationary trends in Myanmar.

ACKNOWLEDGEMENTS

I wish to express my sincere appreciation to Professor Dr. Tin Tin Htwe, the Rector of Yangon University of Economics, for granting me permission to pursue this thesis. Her support and approval have been main in enabling me to start on this thesis endeavor. I am truly grateful for her confidence in my thesis and for the opportunities she has provided me with.

I would also like to extend my gratitude to Professor Dr. Kyaw Tun Naing, the Pro-Rector of Yangon University of Economics, for granting me the approval to conduct this thesis. His recommendation and support have been instrumental in facilitating my thesis endeavors. I am sincerely thankful for her trust and confidence in my work.

I am proudly grateful to Professor Dr. Aye Thida, who serves as the Program Director, Head of Department of Statistics at Yangon University of Economics. Her insightful recommendations and gracious permission to undertake this study have been invaluable to me. I am deeply indebted to her for her support and encouragement throughout this thesis endeavor.

I am deeply thankful to Professor Dr. Hlaing Hlaing Moe, the Head of Department of Applied Statistics at Yangon University of Economics for her meticulous evaluation of our coursework. Her constructive feedback and valuable suggestions have been instrumental in shaping our understanding of statistical methods and their applications in real-world scenarios.

I am sincerely grateful to Daw Khin Nu Win, Associate Professor (Retd.) at the Department of Statistics at Yangon University of Economics, for her continuous support and timely advice. Her helpful comments and kind encouragement have greatly contributed to improving the quality and clarity of this research. I appreciate her generous spirit and willingness to assist whenever I needed help.

I would also like to express my heartfelt thanks to Associate Professor Dr. Khin Thet Tun, from the Department of Statistics at Yangon University of Economics, for her patient guidance and technical assistance throughout this study. Her clear explanations and thoughtful feedback helped me to overcome many academic challenges during the thesis process. Her encouragement has been a constant source of motivation and strength.

I am deeply grateful to Dr. Win Naing Oo, Associate Professor at the Department of Applied Statistics, Yangon University of Economics. His insightful suggestions, expert guidance, and unwavering support have significantly enhanced the quality of my thesis. His mentorship has had a profound impact on my academic journey, and I sincerely appreciate her invaluable contributions to the success of this study.

I am thankful for all professors, associate professor and external lecturer of MAS program who have given me tremendous knowledge throughout the years. And I would like to say my heartiest thanks, all my family, friends, and colleagues who give me the strength and encouragement to accomplish my MAS goal.

CONTENTS

		Page
ABSTRACT		**i**
ACKNOWLEDGEMENTS		**ii**
CONTENTS		**iv**
LIST OF TABLES		**vi**
LIST OF FIGURES		**vii**
LIST OF ABBREVIATIONS		**viii**
CHAPTER I	**INTRODUCTION**	
	1.1 Rationale of the Study	2
	1.2 Objectives of the Study	4
	1.3 Method of Study	4
	1.4 Scope and Limitations of the Study	5
	1.5 Organization of the Study	5
CHAPTER II	**LITERATURE REVIEW**	
	2.1 Overview of Gold Market and CPI in Myanmar	6
	2.2 Gold Market in Emerging Economies	6
	2.3 Consumer Price Index (CPI) in Myanmar	6
	2.4 Gold as an Inflation Hedge: Evidence from Emerging Economies	7
	2.5 Econometric Approaches to Modeling Gold Prices and CPI	8
	2.6 Challenges of Myanmar's Gold Market and CPI	9
	2.7 Policy Implications and the Role of the Central Bank	10
	2.8 Review on Related Studies	11
CHAPTER III	**METHODOLOGY**	
	3.1 Source of Data	14
	3.2 Unit Root Tests	14
	3.3 Setting the Optimal Lag Length of the Model	17
	3.4 Johansen Co-integration Test	18
	3.5 Vector Autoregressive Model	19
	3.6 Lagrange Multiplier (LM) Test for Autocorrelation	21

	3.7	Granger Causality Test	22
CHAPTER IV	**ANALYSIS OF THE RELATIONSHIP BETWEEN 24K GOLD PRICE AND CONSUMER PRICE INDEX**		
	4.1	Descriptive Statistics of 24 Carat Gold Price and Inflation in Myanmar	24
	4.2	Testing the Stationarity of 24k Gold Price and CPI in Myanmar	27
	4.3	VAR Lag Length Selection of 24k Gold Price and CPI in Myanmar	29
	4.4	Johansen Co-integration Test	30
	4.5	Specification and Estimation of VAR Model	30
	4.6	Stability Analysis of VAR Model Eigenvalues	32
	4.7	Summary Statistics and Diagnostics of Residuals	33
	4.8	Lagrange-Multiplier Test for Autocorrelation	33
	4.9	Causality between 24k Gold Price and CPI in Myanmar	34
CHAPTER V	**CONCLUSION**		
	5.1	Findings	36
	5.2	Suggestions and Recommendations	37
	5.3	Further Study	38
	REFERENCES		
	APPENDIX		

LIST OF TABLES

Table No.	Title	Page
3.1	Critical Values for the Dickey–Fuller Test	16
4.1	Descriptive Statistics of 24 Carat Gold Price and CPI in Myanmar from April 2012 to July 2022	26
4.2	Augmented Dickey Fuller and Phillips Perron Tests at Level	28
4.3	Augmented Dickey Fuller and Phillips Perron Tests at First Difference	28
4.4	Lag Length Selection of 24k Gold Price and CPI in Myanmar	29
4.5	Johansen Co-integration Test	30
4.6	VAR Estimation Result	31
4.7	Eigenvalue Stability Check	32
4.8	Error Distribution and Diagnostic Summary	33
4.9	Lagrange-Multiplier Test	34
4.10	Results of Causality Between 24k Gold Price and CPI	35

LIST OF FIGURES

Figure No.	Title	Page
4.1	Gold Price (Kyats per Ounce) in Myanmar	24
4.2	CPI in Myanmar	25

LIST OF ABBREVATIONS

ADB	Asian Development Bank
ADF	Augmented Dickey-Fuller Test
AIC	Akaike Information Criterion
ARCH	Autoregressive Conditional Heteroskedasticity
ARDL	Autoregressive Distributed Lag
CBM	Central Bank of Myanmar
COVID-19	Coronavirus Disease 2019
CPI	Consumer Price Index
CSO	Central Statistical Organization
DF	Dickey-Fuller Test
FPE	Final Prediction Error
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
GDP	Gross Domestic Product
HQ	Hannan-Quinn Criterion
I(0)	Integrated of Order Zero
I(1)	Integrated of Order One
IMF	International Monetary Fund
LM	Lagrange Multiplier
MMK	Myanmar Kyat (Currency)
PP	Phillips-Perron Test
SBC	Schwarz Bayesian Criterion
SVAR	Structural Vector Autoregression
UNDP	United Nations Development Programme
VAR	Vector Autoregression
VECM	Vector Error Correction Model

CHAPTER I
INTRODUCTION

The relationship between gold prices and the Consumer Price Index (CPI) has long been a subject of empirical and theoretical interest in both financial and economic research. Gold is widely regarded as a hedge against inflation due to its intrinsic value and limited supply. Historically, during periods of inflationary pressure and economic uncertainty, investors tend to shift their portfolios toward gold to preserve real purchasing power (Baur & Lucey, 2010; Capie et al., 2005). In contrast, the CPI is a core macroeconomic indicator that tracks the average change over time in the prices paid by consumers for goods and services. It serves not only as a measure of inflation but also as a foundation for central bank policies, wage adjustments, and fiscal planning.

Understanding the dynamic interplay between gold prices and CPI is vital for multiple stakeholders, including policymakers seeking to manage inflation, investors developing risk mitigation strategies, and researchers exploring asset price behavior in response to macroeconomic variables. Prior studies in developed and emerging markets have yielded mixed evidence regarding the strength and direction of this relationship. Some have found that gold acts as an effective long-term hedge against inflation, while others argue that this property is time- and context-dependent (Ghosh et al., 2004; Wang & Lee, 2011). These discrepancies highlight the importance of conducting country-specific research, particularly in economies that are structurally unique or undergoing transition.

In Myanmar, the need to investigate the connection between gold prices and consumer price movements has become increasingly relevant amid recent macroeconomic turbulence, including currency devaluation, global commodity price fluctuations, and ongoing domestic political instability since 2021. Gold holds a significant role in Myanmar's economy, both as a culturally valued asset and as a commonly accepted medium of exchange in informal markets, where confidence in formal financial institutions is often limited (Turnell, 2011). At the same time, changes in the Consumer Price Index (CPI), influenced by factors such as exchange rate swings and supply chain disruptions, directly affect the cost of living and household purchasing capacity (ADB, 2022). Although gold is deeply woven into the country's financial and

cultural practices, the relationship between gold prices and consumer prices in Myanmar remains largely underexamined, leaving a gap in understanding how these two variables interact in the local context.

This study aims to fill this research gap by examining the causal and dynamic relationship between gold prices and CPI in Myanmar using advanced time series econometric methods, including Vector Autoregression (VAR), Granger causality, and impulse response analysis. By analyzing monthly data from 2016 to 2024, capturing both relatively stable and crisis periods this research seeks to uncover whether movements in 24 Carat (24k) gold prices precede changes in CPI or whether inflationary pressures drive gold price adjustments. The outcomes will contribute to the broader literature on CPI hedging in developing economies and offer policy-relevant insights for monetary authorities, investors, and financial planners in Myanmar.

1.1 Rationale of the Study

Gold has historically been regarded as a safe-haven asset, especially during periods of economic uncertainty, high inflation, or currency instability (Baur & McDermott, 2010; World Gold Council, 2020). Globally, investors often turn to gold in times of geopolitical tension, market volatility, or financial crises, as seen during events such as the 2008 global financial crisis and the COVID-19 pandemic (Cheema et al., 2020; Wang et al., 2021). In developed economies, gold is typically part of a diversified investment portfolio, used as a hedge against inflation and currency risks (Baur & Lucey, 2010). Central banks also maintain gold reserves to support monetary stability, highlighting gold's enduring strategic importance in the global financial system (World Gold Council, 2020).

Across Asia, the role of gold is even more pronounced. In countries like India, China, and Vietnam, gold serves not only as a financial instrument but also as a culturally embedded store of wealth, often passed down across generations (World Gold Council, 2019; Asian Development Bank, 2020). These societies exhibit a strong preference for gold due to historical distrust in fiat currencies and underdeveloped financial systems, especially among rural populations (Dempster, 2006; World Gold Council, 2021). In Southeast Asia, gold markets are active, informal, and deeply integrated into household-level financial behavior (Asian Development Bank, 2020; WGC, 2021). Many Asian economies, particularly those with less developed capital

markets or volatile political climates, regard gold as an accessible, liquid, and reliable asset (Baur et al., 2014; World Gold Council, 2021).

In Myanmar, gold occupies a unique and prominent position in the informal economy. It is widely traded not only as a traditional asset but also as a hedge against rapid currency depreciation, political instability, and weak institutional trust. With a fragmented banking sector and limited public access to formal investment products, gold has long been considered a more secure and tangible store of value. Even in the absence of modern financial infrastructure, gold is used for savings, dowries, intergenerational wealth transfers, and emergency funds. Consequently, fluctuations in gold prices are often perceived by the public as a barometer of broader economic sentiment.

In contrast, the Consumer Price Index (CPI) acts as a key gauge of inflation, widely used by policymakers and economists to understand shifts in the cost of living and to steer monetary actions (Investopedia, 2023). As CPI rises, inflationary pressures mount eroding household purchasing power, increasing business expenses, and shaping investment decisions. During such times, households and investors in Myanmar and across Asia frequently turn to gold, perceiving it as a more stable asset amid currency erosion and economic uncertainty (Chua & Woodward, 1982; Vietnam case). This trend is particularly pronounced in regions with high inflation expectations, low financial inclusion, and limited trust in official data or monetary institutions, where gold provides a tangible and accessible alternative.

Understanding the dynamic relationship between gold prices and the Consumer Price Index (CPI) is therefore essential for multiple stakeholders. For investors, insights into this relationship can guide portfolio diversification, risk management, and strategies aimed at countering inflation and interest rate fluctuations (Reuters, 2024). For policymakers and central banks, a clearer grasp of how gold prices interact with inflation and currency perceptions can provide additional tools for assessing inflation expectations and crafting appropriate monetary responses. Recent surveys indicate that central banks are increasingly turning to gold as part of their inflation risk management, with many planning to increase gold reserves in the coming years (World Gold Council, 2024). In economies like Myanmar, where informal markets dominate and financial data is limited, such insights are crucial to bridging the gap between formal economic indicators and real-world financial behavior.

Myanmar's economic landscape is characterized by volatile currency movements, fragmented financial systems, and a strong informal sector. Gold transactions are often conducted outside formal financial institutions, making them difficult to track through official channels. Yet this informality underscores the real-world importance of gold in household financial decisions. Persistent inflation, exchange rate instability, political uncertainty, and limited access to banking services mean that traditional models of inflation expectations may not fully capture Myanmar's unique environment (World Bank, 2024). As such, studying the gold–CPI relationship in this context provides an opportunity to generate insights that complement international findings while addressing localized financial behaviors.

From this study, the primary objective is to examine the casual relationship between the 24K gold price and the Consumer Price Index (CPI) in the Myanmar. While the link between gold and inflation may appear intuitive, the specific dynamics, direction, and strength of this relationship in Myanmar remain largely unquantified and underexplored. This study seeks to provide empirical evidence on whether changes in CPI can predict movements in gold prices, or conversely, whether gold prices can signal shifts in consumer prices, using causality tests and time series analysis.

By focusing directly on this relationship, the research aims to contribute a focused and context-specific finding that can be practically useful for investors, households, and policymakers. The outcome of this study may help improve understanding of how gold functions within Myanmar's unique economic environment, where traditional financial indicators often fail to fully capture real-world behaviors. In doing so, this research aspires to offer concrete evidence that can inform financial decisions, investment strategies, and potential areas for further policy investigation in Myanmar and other similar developing economies.

1.2 Objectives of the Study

The objectives of this study are as follows:

(i) To describe the trends and patterns in 24k gold prices and CPI in Myanmar.

(ii) To investigate the causal relationship between 24k gold prices and CPI in Myanmar.

1.3 Method of Study

Firstly, descriptive statistics was used to summarize the 24k gold prices and the Consumer Price Index (CPI) in Myanmar. Secondly, a unit root test was conducted to assess the stationarity of the data, and Johansen co-integration test for long run relationship among variables were used. Finally, Vector Autoregressive (VAR) Model was applied to investigate whether a causal relationship existed between 24k gold prices and the Consumer Price Index (CPI).

1.4 Scope and Limitations of the Study

This study examines the secondary monthly data on 24k gold prices (in MMK per ounce) and CPI from April 2012 to July 2022, sourced from Selected Monthly Economic Indicator of the Central Statistical Organization (CSO). This study is limited to other macroeconomic variables such as exchange rates and interest rates that may influence gold prices and CPI. Additionally, limitation includes external factors such as political instability and global gold market fluctuations.

1.5 Organization of the Study

The study is structured into five chapters to systematically explore the relationship between gold prices and the Consumer Price Index (CPI) in Myanmar. Chapter 1 provides the background and rationale of the study. Chapter 2 examines existing theoretical and empirical studies on the gold-CPI relationship. Chapter 3 emphasizes methodology about the econometric models and tests used for analysis. Chapter 4 presents interpreting the results of stationarity tests, cointegration test, model results and causality to determine whether gold prices influence CPI or vice versa. Finally, Chapter 5 summarizes the key findings, discusses their implications for investors and policymakers, and suggests future research.

CHAPTER II
LITERATURE REVIEW

This chapter explores the theoretical foundations and empirical studies concerning the relationship between gold prices and Consumer Price Index (CPI), with a particular focus on Myanmar's economic context. The behavior of gold markets and Consumer Price Index (CPI) in Myanmar is examined, followed by econometric approaches to study between gold prices and CPI. This chapter also examines challenges in Myanmar's gold market and CPI, along with a discussion on policy implications including the role of the Central Bank of Myanmar (CBM), and a review of related studies is also included.

2.1 Overview of Gold Market and CPI in Myanmar

Gold markets play a vital role in financial stability, inflation hedging, and wealth preservation, particularly in emerging economies. In Myanmar, 24-carat gold has long been considered a safe-haven asset amid economic uncertainties and inflationary pressures. The Consumer Price Index (CPI) measures the average change over time in the prices paid by consumers for a basket of goods and services, serving as a key indicator of inflation. Understanding the relationship between gold prices and Consumer Price Index (CPI), is crucial for investors, policymakers, and economists (Baur & Lucey, 2010).

2.2 Gold Market in Emerging Economies

Emerging markets like Myanmar exhibit unique financial dynamics, including high inflation volatility, currency depreciation, and political instability all of which influence gold prices. According to Bekaert and Harvey (2017), in such economies, gold often acts as a store of value and a hedge against inflation and currency risk. Aung and Khaing (2021) noted that in Myanmar, 24-carat gold demand has remained resilient due to low trust in financial institutions, limited access to banking services, and macroeconomic instability. Despite informal trading channels and limited regulation, gold remains a critical financial asset for households and investors.

2.3 Consumer Price Index (CPI) in Myanmar

The Consumer Price Index (CPI) serves as a key indicator of price trends, reflecting changes in the average cost of a basket of goods and services consumed by households. In Myanmar, the CPI is one of the primary tools used to monitor changes in consumer prices; however, the accuracy and frequency of CPI reporting have been affected in recent years by political disruptions and limited institutional capacity. Price fluctuations in Myanmar have often been influenced by factors such as currency movements, import costs, and supply-side challenges, especially in essential goods like food and fuel (IMF, 2022). Since the 2021 political crisis, these pressures have grown more pronounced, contributing to volatility in household purchasing power. As a trusted store of value, gold continues to play a key role in wealth preservation, especially during periods of economic uncertainty, creating a linkage between CPI movements and gold market behavior. Understanding this relationship requires accounting for structural shifts, data constraints, and informal market dynamics (Turnell, 2022).

2.4 Gold as an Inflation Hedge: Evidence from Emerging Economies

A substantial body of literature has investigated the relationship between gold prices and inflation, particularly in the context of emerging markets where economic volatility often prompts the public to rely on alternative stores of value. In Vietnam, Nguyen and Tran (2020) identified a positive correlation between inflation and gold prices, attributing this link to investors' behavior during inflationary spikes. Their study revealed that as consumer prices rise, the demand for gold as a hedge increases, reinforcing gold's traditional role as a safe asset in times of economic distress.

Similarly, in India, Jain (2012) found strong evidence supporting the role of gold as a long-term inflation hedge, particularly during periods of monetary instability and high inflation expectations. The study emphasized the cultural and economic relevance of gold in Indian households, highlighting how gold purchases intensify in response to eroding purchasing power and distrust in fiat currency. These findings resonate with other research from emerging markets, which suggests that the efficacy of gold as an inflation hedge often depends on institutional quality, access to formal finance, and cultural preferences.

In contrast, the academic literature on Myanmar remains limited. However, preliminary evidence indicates a similar trend. A report by JETRO (2013) observed that

rising CPI levels in Myanmar appear to be associated with increased 24-carat gold prices, largely driven by diminished public trust in the local currency and a rising demand for tangible, tradable assets. The informal nature of Myanmar's gold market and the widespread use of gold as a store of value in daily life further emphasize the need for focused empirical investigation in this under-researched context.

2.5 Econometric Approaches to Modeling Gold Prices and CPI

Analyzing the interplay between gold prices and CPI requires the use of advanced time series econometric methods, particularly in environments where high-frequency and reliable data are scarce (Enders, 2015). Time series models not only help in capturing trends and patterns over time but also offer tools to examine causal relationships and dynamic interdependencies between macroeconomic variables. A foundational step in such analyses involves testing for stationarity using tools like the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests. These tests determine whether time series data exhibit unit roots, a critical prerequisite for avoiding spurious regression results and ensuring model validity (Dickey & Fuller, 1979).

To assess the dynamic relationship between variables, the Vector Autoregression (VAR) model is widely used. VAR captures bidirectional interactions over time without requiring prior assumptions about causality (Sims, 1980). Within the VAR framework, the Granger causality test is particularly useful for identifying whether past values of one variable, such as CPI, can statistically predict future movements in another variable, such as gold prices (Granger, 1969). These techniques have been effectively applied in other data-constrained settings. Musonera (2020) employed VAR and Granger causality analyses across several African economies, demonstrating their robustness in capturing asset price dynamics even with limited datasets. His findings suggest that these tools are adaptable and valuable in contexts like Myanmar, where data availability is often limited, but economic volatility necessitates insightful forecasting models.

Applying these econometric methods to Myanmar's inflation and gold market data can thus offer credible and policy-relevant insights, especially when formal financial indicators are insufficient to understand household behavior and market expectations. This methodological approach enables researchers and policymakers to

better navigate informal financial systems, assess market sentiment, and design interventions rooted in empirical analysis.

2.6 Challenges of Myanmar's Gold Market and CPI

One of the major obstacles in empirically analyzing the relationship between gold prices and the Consumer Price Index (CPI) in Myanmar is the limited availability and reliability of historical data. Official statistical releases have not updated CPI or gold price figures beyond July 2022, leaving a critical gap for contemporary time series analysis (CSO, 2022). Moreover, earlier datasets often suffer from inconsistencies due to changes in data collection methodologies, shifts in CPI base years, and the absence of standardized reporting protocols (ADB, 2021). These data quality concerns constrain the robustness of time series models such as Vector Autoregression (VAR) and Vector Error Correction Models (VECM), both of which require long-term, high-frequency, and internally consistent datasets for accurate model estimation (Enders, 2014).

Myanmar's macroeconomic landscape has been marked by significant structural and political volatility, particularly since the events of 2021. These disruptions contribute to price instability and complicate the estimation of stable relationships between macroeconomic indicators such as CPI and gold prices. Unpredictable movements in the exchange rate and inconsistent policy communication further cloud market expectations, thereby increasing the likelihood of structural breaks and non-linear dynamics in the data (World Bank, 2023). Such conditions reduce the effectiveness of conventional time series models, as their assumptions of linearity and stable relationships may no longer hold in practice (Gujarati & Porter, 2009).

A further complication arises from the dominance of informal gold trading in Myanmar. A substantial share of gold transactions takes place outside the formal financial system, especially in rural regions and through unlicensed dealers (OECD, 2022). These transactions are rarely captured in official statistics, leading to discrepancies between published gold prices and actual market behavior. This disconnect limits the reliability of empirical findings and weakens the applicability of gold price data as an explanatory variable in relation to CPI. In addition, the informal nature of the gold market undermines regulatory oversight and impedes the development of policy measures designed to monitor or influence price behavior

(UNESCAP, 2021). Consequently, researchers must rely on proxy data or unverified sources, which introduces uncertainty into the analysis and may reduce the credibility of conclusions.

2.7 Policy Implications and the Role of the Central Bank

The Central Bank of Myanmar (CBM) holds a pivotal role in shaping monetary conditions that affect both CPI trends and domestic gold market dynamics. Gold in Myanmar functions not only as a store of value but also as a parallel monetary asset, especially when there is reduced confidence in the national currency. The CBM's adoption of a managed float exchange rate system has had direct consequences for domestic gold pricing, as shifts in the kyat's value can significantly alter local gold prices even when international gold prices remain stable (CBM, 2022). Through mechanisms such as currency auctions and foreign exchange interventions, the CBM indirectly influences the gold market by affecting gold affordability and supply conditions.

Monetary policy decisions particularly those involving interest rates also play a role in shaping consumer behavior and asset allocation. In Myanmar's context, where formal savings instruments are less prevalent and financial intermediation remains underdeveloped, households often seek alternative means to maintain the purchasing power of their assets, with gold being a popular option (IMF, 2021). This behavior, although rational at the household level, creates challenges for monetary authorities aiming to anchor price expectations and maintain a stable CPI trend. Transparent, credible, and timely policy communication by the CBM is therefore essential in shaping public expectations and reducing excessive shifts in demand between monetary assets and commodities like gold.

Beyond conventional policy tools, the CBM working in coordination with relevant ministries and financial institutions can take active steps to formalize the gold market and strengthen its integration into the regulated financial sector. For instance, launching public awareness campaigns that educate consumers about CPI, price behavior, and the benefits of formal financial instruments can encourage diversification of household savings (GIZ, 2022). Additionally, improving the legal and institutional

framework for gold trading by introducing licensing requirements, standardizing price reporting, and enhancing traceability of gold flows would help reduce market fragmentation. These reforms would not only enhance the effectiveness of monetary policy but also support more accurate macroeconomic modeling and forecasting by improving the quality and coverage of gold price data (UNDP, 2022).

2.8 Review on Related Studies

Narayan et al. (2010) analyzed the influence of inflation and interest rates on gold prices using data from the United States over the period 1970 to 2008. The study applied the autoregressive distributed lag (ARDL) bounds testing approach to examine the existence of a long-run cointegration relationship among gold prices, inflation, and interest rates. Additionally, Granger causality tests were conducted to determine the direction of short-run dynamics. The results revealed that inflation Granger-caused gold prices, supporting the role of gold as an inflation hedge. However, in the short run, the relationship was less stable. The study emphasized the importance of monetary policy consistency in mitigating the inflationary impact on gold demand and prices.

Baur and Lucey (2010) examined the role of gold as a safe-haven asset during financial crises using daily data from 1995 to 2009. Applying GARCH models and correlation analysis, they found that gold acts as a safe haven against stocks but only in extreme market conditions. The study revealed that gold's hedging capability is time-varying, with stronger effects during periods of high volatility. The results suggested that investors should consider gold as a diversification tool primarily during financial turmoil rather than in stable market conditions.

Wang et al. (2011) investigated the relationship between gold prices, oil prices, and the U.S. dollar index using weekly data from 2000 to 2010. Employing a VAR model and Granger causality tests, they found that oil prices and the dollar index significantly influenced gold prices. The impulse response analysis indicated that a shock to oil prices led to a prolonged increase in gold prices, while a stronger dollar suppressed them. The study concluded that gold prices are driven by both macroeconomic fundamentals and speculative demand.

Sujit and Kumar (2011) studied the relationship between gold price, crude oil price, and the Indian rupee to US dollar exchange rate using monthly data from January 2007 to June 2011. The study aimed to understand the dynamic interactions among these variables by employing the Vector Autoregression (VAR) model, which allowed the researchers to capture the interdependencies without imposing a priori causality. The model was supplemented with impulse response functions (IRFs) and variance decomposition analysis to trace the effect of shocks in one variable on the others over time. The findings indicated that gold prices were significantly influenced by fluctuations in the exchange rate and oil prices. The impulse response functions showed that a shock to the exchange rate led to a considerable and persistent increase in gold prices. The study concluded that policymakers should consider external price shocks in managing domestic inflation and currency stability.

Dimitriou and Simos (2013) explored the causal relationship between gold prices and consumer price indices in the United States and the Eurozone using monthly data from 1999 to 2011. Employing Johansen cointegration analysis and a VECM model, the study confirmed a long-run equilibrium relationship between gold prices and inflation. The Granger causality test within the VECM framework indicated the relationship between gold prices and CPI in the U.S., while in the Eurozone, only CPI Granger-caused gold prices. The findings reinforced the hypothesis that gold serves as a hedge against inflation, particularly in developed economies with transparent monetary policy.

Beckmann and Czudaj (2013) analyzed whether gold serves as an inflation hedge in major economies (U.S., U.K., Japan, and the Eurozone) from 1975 to 2011. Using a time-varying cointegration approach, they found that gold's inflation-hedging ability varies across periods and countries. The results showed that gold was a strong hedge in the U.S. and U.K. but weaker in Japan. The study emphasized that gold's effectiveness as an inflation hedge depends on monetary policy regimes and inflation expectations.

Tiwari et al. (2015) studied the dynamic relationship between gold prices, inflation, and exchange rates in India using monthly data from 1990 to 2014. Applying wavelet coherence and nonlinear causality tests, they found that gold prices and

inflation were strongly connected in the long run, while exchange rate fluctuations had short-term impacts. The study highlighted that gold acts as a hedge against inflation and currency depreciation in India, particularly during economic uncertainty.

Salisu et al. (2020) explored the hedging and safe-haven properties of gold against global economic policy uncertainty (EPU) from 1997 to 2018. Using quantile regression and causality-in-quantiles approaches, they found that gold serves as both a hedge and a safe haven against EPU shocks. The results indicated that gold's hedging effectiveness increases during extreme uncertainty episodes. The study recommended gold as a strategic asset for risk-averse investors during periods of heightened policy uncertainty.

Zin (2022) investigated the relationship between gold prices and the exchange rate in Myanmar from 2011 to 2020. Monthly data was obtained from the Central Statistical Organization of Myanmar. Techniques used include unit root tests, Vector Autoregressive (VAR) lag length of the model, Johansen cointegration test, and Granger causality test. The study found that both gold prices and exchange rates are stationary at the first difference. The Johansen cointegration test revealed a long- run relationship between the variables. The Granger causality test showed a unidirectional causal relationship from the exchange rate to gold prices. The study concluded that changes in the exchange rate impact gold prices in Myanmar.

Aung (2023) analyzed the cointegration and causal relationship between export and import in ASEAN countries from 1970 to 2021. Annual trade data was obtained from the United Nations Database. The study applied the Augmented Dickey-Fuller test, Engle-Granger Cointegration Test, Johansen Cointegration Test, Vector Error Correction Model (VECM), and Vector Autoregressive (VAR) Model. The findings showed cointegration for Singapore and Thailand, while Indonesia and Myanmar exhibited unidirectional causality from export to import, and Cambodia and Vietnam had bidirectional causality. The study recommended promoting exports in Indonesia and Myanmar to support import demand.

CHAPTER III
METHODOLOGY

In this chapter, the contents are divided into two parts. The first part explains the sources of data, and the second part presents descriptive statistics, VAR/ VECM model estimation and Granger causality test was used to check the cointegration and causal relationship between among 24k gold price and Consumer Price Index (CPI).

3.1 Source of Data

This study aims at investigating the cointegration and causal relationship between 24k gold price and CPI in Myanmar. In this study uses the monthly average 24k gold price and monthly CPI data obtained from the authenticated sources of Selected Monthly Economic Indicators of Central Statistical Organization (CSO) website for a period of about 10 years from April 2012 to July 2022 with 124 observations.

3.2 Unit Root Tests

Before estimating the long-run relationship, both the variables are tested for their unit root test. Plotting the data and observing the graph is sometimes very useful because it can indicate clearly the presence or not of deterministic regressors. However, this procedure is the most sensible way to test for unit roots when the form of the data-generating process is unknown. In this present study, the Augmented Dickey-Fuller (ADF) Unit root test has been used to examine the stationarity of the data. The ADF Unit Root test is based on the null hypothesis Ho (the series is not stationary). If the calculated ADF statistic is less than the critical value, then the null hypothesis is rejected; otherwise accepted. If the variable is stationary at the first level, the variable is said to be integrated of order zero, I (0). If the variable is non stationary at level, the ADF test can be utilized and the first difference of the variable can be used for testing a unit root. In this case, the variable is said to be co-integrated of order one, I (1).

(i) Augmented Dickey Fuller Test

Dickey and Fuller (1979, 1981) devised a formal procedure to test for non stationarity. The key insight of their test is that testing for non-stationarity is equivalent to testing for the existence of a unit root. Thus the obvious test is the following, which is based on the simple AR (1) model of the form:

$$y_t = \varphi y_{t-1} + u_t \tag{3.1}$$

To examine here is whether φ is equal to 1 (unity and hence 'unit root'). Obviously, the null hypothesis H0: $\varphi = 1$, and the alternative hypothesis H1: $\varphi < 1$. A different (more convenient) version of the test can be obtained by subtracting y t−1 from both sides of Equation

$$y_t - y_{t-1} = (\varphi - 1)\, y_{t-1} + u_t \tag{3.2}$$

$$\Delta y_t = (\varphi - 1)\, yt_{-1} + u_t \tag{3.3}$$

$$\Delta y_t = \gamma\, y_{t-1} + u_t \tag{3.4}$$

where of course $\gamma = (\varphi - 1)$. Now the null hypothesis is H0: $\gamma = 0$ and the alternative hypothesis Ha: $\gamma < 0$, where if $\gamma = 0$ then y_t follows a pure random-walk model. Dickey and Fuller (1979) also proposed two alternative regression equations that can be used for testing for the presence of a unit root. The first contains a constant in the random-walk process, as in the following equation:

$$\Delta y_t = \alpha_0 + \gamma\, y_{t-1} + u_t \tag{3.5}$$

This is an extremely important case, because such processes exhibit a definite trend in the series when $\gamma = 0$ which is often the case for macroeconomic variables. The second case is also to allow a non-stochastic time trend in the model, to obtain:

$$\Delta y_t = \alpha_0 + \beta_{2t} + \gamma\, y_{t-1} + u_t \tag{3.6}$$

The DF test for stationarity is then simply the normal t-test on the coefficient of the lagged dependent variable y_{t-1} from one of the three models. This test does not, however, have a conventional t-distribution and so we must use special critical values originally calculated by Dickey and Fuller.

MacKinnon (1991) tabulated appropriate critical values for each of the three models discussed above and these are presented in Table (3.1).

Table (3.1) Critical Values for the Dickey–Fuller Test

Model	1%	5%	10%
$\Delta y_{t-1} = \gamma\, y_{t-1} + u_t$	−2.56	−1.94	−1.62
$\Delta y_{t-1} = \alpha_0 + \gamma\, y_{t-1} + u_t$	−3.43	−2.86	−2.57
$\Delta y_{t-1} = \alpha_0 + \beta_{2t} + \gamma\, y_{t-1} + u_t$	−3.96	−3.41	−3.13
Standard critical values	−2.33	−1.65	−1.28

Source: MacKinnon (1991)

In all cases, the test focuses on whether $\gamma = 0$. The DF test statistic is the t statistic for the lagged dependent variable. If the DF statistical value is smaller than critical value and the null hypothesis of a unit root is rejected and conclude that y_t is a stationary process.

As the error term is unlikely to be white noise, Dickey and Fuller extended their test procedure by suggesting an augmented version of the test that includes extra lagged terms of the dependent variable in order to eliminate autocorrelation. The lag length on these extra terms is either determined by the Akaike information criterion (AIC) or the Schwartz Bayesian criterion (SBC), or more usefully by the lag length necessary to whiten the residuals (that is after each case we check whether the residuals of the ADF regression are auto correlated or not through a Lagrange multiplier (LM) tests rather than the Durbin-Watson (DW) test. The three possible forms of the ADF test are given by the following equations:

$$\Delta y_t = \gamma\, y_{t-1} + \sum_{i=1}^{p} \beta_i\, \Delta y_{t-i} + u_t \qquad (3.7)$$

$$\Delta y_t = \alpha_0 + \gamma\, y_{t-1} + \sum_{i=1}^{p} \beta_i\, \Delta y_{t-i} + u_t \qquad (3.8)$$

$$\Delta y_t = a_0 + \gamma\, y_{t-1} + a_2 t + \sum_{i=1}^{p} \beta_i\, \Delta y_{t-i} + u_t \qquad (3.9)$$

The difference between the three regressions again concerns the presence of the deterministic elements a_0 and a_{2t}. The critical values for the ADF tests are the same as those given in Table 3.1 for the DF test. Unless the econometrician knows the actual data-generating process, there is a question concerning whether it is most appropriate to estimate equation. Doldado et al. (1990) suggest a procedure which starts from the estimation of the most general model given by Equation, answering a set of questions regarding the appropriateness of each model and then moving to the next model.

(ii) Phillips Perron Test

The distribution theory supporting the DF and ADF tests is based on the assumption that the error terms are statistically independent and have a constant variance. So, when using the ADF methodoLogy, one has to make sure that the error terms are uncorrelated and that they really do have a constant variance. Phillips and Perron (1988) developed a generalization of the ADF test procedure that allows for fairly mild assumptions concerning the distribution of errors. The test regression for the PP test is the AR (1) process:

$$\Delta y_{t-1} = \alpha_0 + \gamma\, y_{t-1} + e_t \qquad (3.10)$$

While the ADF test corrects for higher-order serial correlation by adding lagged differenced terms on the right-hand side, the PP test makes a correction to the t statistic of the coefficient γ from the AR (1) regression to account for the serial correlation in et. So the PP statistics are only modifications of the ADF t-statistics that take into account the less restrictive nature of the error process. The expressions are extremely complex to derive and are beyond the scope of this text. However, since many statistical packages have routines available to calculate these statistics, it is good for the researcher to test the order of integration of a series by also performing the PP test. The asymptotic distribution of the PP t statistic is the same as the ADF t-statistic and therefore the MacKinnon (1991) critical values are still applicable. As with the ADF test, the PP test can be performed with the inclusion of a constant, a constant and a linear trend, or neither in the test regression.

3.3 Setting the Optimal Lag Length of the Model

In economics, the dependent variable (Y) on another variable(s) and the independent variable (X) are rarely instantaneous. Furthermore, Y responds to X with a lapse of time. Such a lapse of time is called a lag. The issue of finding the appropriate (optimal) lag length is very important because we want to have Gaussian error terms that is standard normal error terms that do not suffer from non-normality, autocorrelation, heteroskedasticity and so on.

Setting the value of the lag length is affected by the omission of variables that might affect only the short-run behavior of the model. This is because omitted variables

instantly become part of the error term. Therefore, very careful inspection of the data and the functional relationship is necessary before proceeding with estimation, to decide whether to include additional variables. It is quite common to use dummy variables to take into account short-run 'shocks' to the system, such as political events that had important effects on macroeconomic conditions.

The most common procedure in choosing the optimal lag length is to estimate Vector Autoregressive (VAR) model including all our variables in levels (nondifferenced data). This VAR model should be estimated for a large number of lags, then reducing down by estimating the model for one lag less until zero lags are reached (that is to estimate the model for 12 lags, then 11, then 10 and so on until reach 0 lags). In each of these models inspect the values of the AIC and the SBC criteria, as well as the diagnostics concerning autocorrelation, heteroskedasticity, possible Autoregressive Conditional Hetroskedasticity (ARCH) effects and normality of the residuals. The HQ and SBC criteria impose stronger penalties for overparameterization compared to AIC (Hamilton, 1994, pp. 308-310), making them more reliable for finite samples by reducing the risk of overfitting.

3.4 Johansen Co-integration Test

After unit root or stationary test, it can be found out whether co-integration test is necessary to identify the long run relationship among the variables. If the time series are stationary and integrated of order I (0), co-integration test is not necessary to perform since any shock to the system in the short run will adjust to the long run. If the time series are integrated of order I (1) which means the series become stationary after first difference, performing co-integration test is necessary to check whether the long run relationship exit among the variables. One prominent co-integration test is Johansen co-integration test. Under Johansen co-integration test, the null hypothesis is there is no integration against the alternative hypothesis which is there is co-integration. The decision criteria is that null hypothesis can be rejected if the Trace statistics and maximum Eigenvalue is greater than 5% critical value.

Trace Test

The trace test is a test whether the rank of the matrix is γ_0. The null hypothesis is that rank (Π) = γ_0. The alternative hypothesis is that γ_0 < rank (Π) < n, where n is the maximum number of possible co-integrating vectors. For the succeeding test if this null hypothesis is rejected, the next null hypothesis is that rank (Π) = γ_0 + 1 and the alternative hypothesis is that $\gamma_0 + 1 < rank \leq n$.

The likelihood ratio test statistic is

$$LR(\gamma_0, n) = -T \sum_{i=\gamma_0+1}^{n} \ln(1 - \lambda_i) \tag{3.11}$$

where LR (γ_0, n) is the likelihood ratio statistic for testing whether rank (Π) = r versus the alternative hypothesis that rank (Π) ≤ n. For example, the hypothesis that rank (Π) = 0 versus the alternative that rank (Π) ≤ n is tested by the likelihood ratio test statistic $LR(\gamma_0, n) = -T \sum_{i=\gamma_0+1}^{n} \ln(1 - \lambda_i)$.

Why does the trace test refer to itself as the "trace test"? The trace test gets its name from the fact that the asymptotic distribution of the test statistic is the trace of a matrix based on conventional Wiener processes or functions of Brownian motion (Johansen, 1995).

3.5 Vector Autoregressive Model

It is quite common in economics to have models in which some variables are not only explanatory variables for a given dependent variable, but are also explained by the variables that they are used to determine. In these cases, the models have simultaneous equations, in which it is necessary to identify clearly which are the endogenous and which are the exogenous or predetermined variables.

The decision regarding such a differentiation among variables was heavily criticized by Sims (1980). According to Sims, if there is simultaneity among a number of variables, then all these variables should be treated in the same way. In other words, there should be no distinction between endogenous and exogenous variables. Therefore, once this distinction is abandoned, all variables are treated as endogenous. This means that in its general reduced form each equation has the same set of regressors, which leads to the development of VAR models.

When there are not confident that a variable really is exogenous, each variable has to be treated symmetrically. The time series y_t that is affected by current and past values of x_t and, simultaneously, the time series x_t to be a series that is affected by current and past values of the y_t series. In this case the simple bivariate model is given by:

$$y_t = \beta_{10} - \beta_{12}x_t + \gamma_{11}y_{t-1} + \gamma_{12}x_{t-1} + u_{yt} \quad (3.12)$$

$$x_t = \beta_{20} - \beta_{21}y_t + \gamma_{21}y_{t-1} + \gamma_{22}x_{t-1} + u_{xt} \quad (3.13)$$

where we assume that both y_t and x_t are stationary, and u_{yt} and u_{xt} are uncorrelated white-noise error terms. Equations constitute a first-order VAR model, because the longest lag length is unity. These equations are not reduced-form equations, since y_t has a contemporaneous impact on x_t (given by $-\beta_{21}$), and x_t has a contemporaneous impact on y_t (given by $-\beta_{12}$). Rewriting the system using matrix algebra, we get:

$$\begin{bmatrix} 1 & \beta_{12} \\ \beta_{21} & 1 \end{bmatrix}\begin{bmatrix} \gamma_t \\ X_t \end{bmatrix} = \begin{bmatrix} \beta_{10} \\ \beta_{20} \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix}\begin{bmatrix} \gamma_{t-1} \\ X_{2t-1} \end{bmatrix} + \begin{bmatrix} u_t \\ u_{Xt} \end{bmatrix} \quad (3.14)$$

$$Bz_t = \Gamma_0 + \Gamma_1 z_{t-1} + u_t \quad (3.15)$$

$$B = \begin{bmatrix} 1 & \beta_{12} \\ \beta_{21} & 1 \end{bmatrix}, z_t = \begin{bmatrix} \gamma_t \\ X_t \end{bmatrix} \Gamma_0 = \begin{bmatrix} \beta_{10} \\ \beta_{20} \end{bmatrix} \quad (3.16)$$

$$\gamma_1 = \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix}, \text{and } u_t = \begin{bmatrix} u_t \\ u_{Xt} \end{bmatrix} \quad (3.17)$$

Multiplying both sides by B^{-1} we obtain: $z_t = A_0 + A_1 z_{t-1} + e_t$ (3.18)

where $A_0 = B^{-1}\Gamma_0$, $A_1 = B^{-1}\Gamma_1$, and $e_t = B^{-1}u_t$. For purposes of notational simplification can denote as a_{i0}; the ith element of the vector A_0; a_{ij} the element in row i and column j of the matrix A_1; and e_{it} as the ith element of the vector e_t. Using this, we can rewrite the VAR model as:

$$y_t = a_{10} + a_{11}y_{t-1} + a_{12}x_{t-1} + e_{1t} \quad (3.19)$$

$$x_t = a_{20} + a_{21}y_{t-1} + a_{22}x_{t-1} + e_{2t} \quad (3.20)$$

To distinguish between the original VAR model and the system have just obtained, the first a structural or primitive VAR system and the second a VAR in standard (or reduced) form. It is important to note that the new error terms, e_{1t} and e_{2t}, are composites of the two shocks u_{yt} and u_{xt}. Since $e_t = B-1\ u_t$ we can obtain e_{1t} and e_{2t} as:

$$e_{1t} = \frac{(u_{yt} + \beta_{12}u_{xt})}{(1 - \beta_{12}\beta_{21})} \tag{3.21}$$

$$e_{2t} = \frac{(u_{xt} + \beta_{21}u_{yt})}{(1 - \beta_{12}\beta_{21})} \tag{3.22}$$

Since u_{yt} and u_{xt} are white-noise processes, follows that both e_{1t} and e_{2t} are also white-noise processes.

The VAR model approach has some very good characteristics. First, it is very simple. The econometrician does not have to worry about which variables are endogenous or exogenous. Second, estimation is also very simple, in the sense that each equation can be estimated separately with the usual OLS method. Third, forecasts obtained from VAR models are in most cases better than those obtained from the far more complex simultaneous equation models (Mahmoud, 1984; McNees, 1986).

However, on the other hand, VAR models have faced severe criticism over various points. First, they are atheoretic, in that they are not based on any economic theory. Since initially there are no restrictions on any of the parameters under estimation, in effect 'everything causes everything'. However, statistical inference is often used in the estimated models so that some coefficients that appear to be insignificant can be dropped, in order to lead to models that might have an underlying consistent theory. Such inference is normally carried out using what are called causality tests.

3.6 Lagrange Multiplier (LM) Test for Autocorrelation

Time series data, comprising observations recorded in a sequential order over time, often violate the classical regression assumption of uncorrelated residuals. According to this assumption, there should be no consistent pattern or correlation between residuals at various time intervals, indicating that the error terms should be independent across time. Autocorrelation is an issue that occurs when this assumption is broken. Autocorrelation affects the efficiency of the least squares estimators, resulting in incorrect standard errors and inference, but it does not skew or cause inconsistency. The Lagrange Multiplier (LM) test is frequently used to find autocorrelation in the residuals of time series models like Vector Autoregression (VAR)

or Vector Error Correction Models (VECM). A General Test of Autocorrelation: Lagrange Multiplier (LM) test proceeds as follows:

(i) Regress on the restricted set of independent variables and save the residuals, $\hat{u}$

(ii) Regress $\hat{u}$ on all of the independent variables and obtain the R-squared, and $\hat{u}_{t-1}$, $\hat{u}_{t-1}$, $\hat{u}_{t-2}$, …,$\hat{u}_{t-p}$, where the latter are the lagged values of the estimated residuals in step 1. Thus, if p = 4, it introduces four lagged values of the residuals as additional regressors in the model. Note that to run this regression it has only (n − p) observations (why?). In short, run the following regression:

$$\hat{u}_t = \alpha_1 + \alpha_2 X_t + \hat{p}_1\hat{u}_{t-1} + \hat{p}_2\hat{u}_{t-2}, + \cdots + \hat{p}_p\hat{u}_{t-p} + \varepsilon_t \quad (3.23)$$

and obtain R^2 from this (auxiliary) regression

(iii) If the sample size is large (technically, infinite), Breusch and Godfrey have shown that

$$LM = (n-p)R^2 \quad (3.24)$$

obtained from the auxiliary regression follows asymptotically a chi-square distribution with p degrees of freedom, i.e.,

$$LM = (n-p)R^2 \chi_p^2 \quad (3.25)$$

where, n is the sample size, p is the number of lagged residuals, R^2 is the coefficient of determination from the auxiliary regression of residuals.

The Lagrange Multiplier (LM) statistic is a test statistic that quantifies the degree of autocorrelation up to lag p. The null hypothesis that there is no autocorrelation is rejected if the computed LM statistic is greater than the crucial value from the chi-square distribution at the selected significance level. This suggests that there is serial correlation in the residuals as at least one of the lagged residual coefficients, ρ, is substantially different from zero.

Test hypothesis is

H_0 : No autocorrelation in residuals.

H_1 : Autocorrelation exists at one or more lags.

3.7 Granger Causality Test

One of the good features of VAR models is that they allow us to test for the direction of causality. Causality in econometrics is somewhat different from the concept

in everyday use; it refers more to the ability of one variable to predict (and therefore cause) the other. Suppose two variables, say y_t and x_t, affect each other with distributed lags. The relationship between these variables can be captured by a VAR model. In this case it is possible to state that (a) y_t causes x_t (b) x_t causes y_t (c) there is a bidirectional feedback (causality among the variables) (d) the two variables are independent. The problem is to find an appropriate procedure that allows us to test and statistically detect the cause and effect relationship among the variables.

Granger (1969) developed a relatively simple test that defined causality as follows: a variable y_t is said to Granger cause x_t if x_t can be predicted with greater accuracy by using past values of the y_t variable rather than not using such past values, all other terms remaining unchanged. The next section presents the Granger causality test, and this will be followed by an alternative causality test developed by Sims (1972).

The Granger causality test for the case of two stationary variables y_t and x_t involves as a first step the estimation of the following VAR model:

$$y_t = a_1 + \sum_{i=0}^{n} \beta_i x_{t-i} + \sum_{j=0}^{m} \gamma_j y_{t-j} + e_{1t} \tag{3.26}$$

$$x_t = a_2 + \sum_{i=0}^{n} \theta_i x_{t-i} + \sum_{j=0}^{m} \delta_j y_{t-j} + e_{2t} \tag{3.27}$$

where it is assumed that both ε_{yt} and ε_{xt} are uncorrelated white-noise error terms.

Granger causality test of hypotheses are as follows:

H_0: X does not Granger cause Y

H_1: X Granger causes Y

Since the probabilities of all different periods are more the 5% then there is no Granger causality between the two variables. To underline the causal direction between the variables, a pairwise Granger causality test has been conducted. In testing the pairwise Granger causality, at least two variables are usually analyzed together while testing for their interaction.

Following are the possible results of the analyzes:

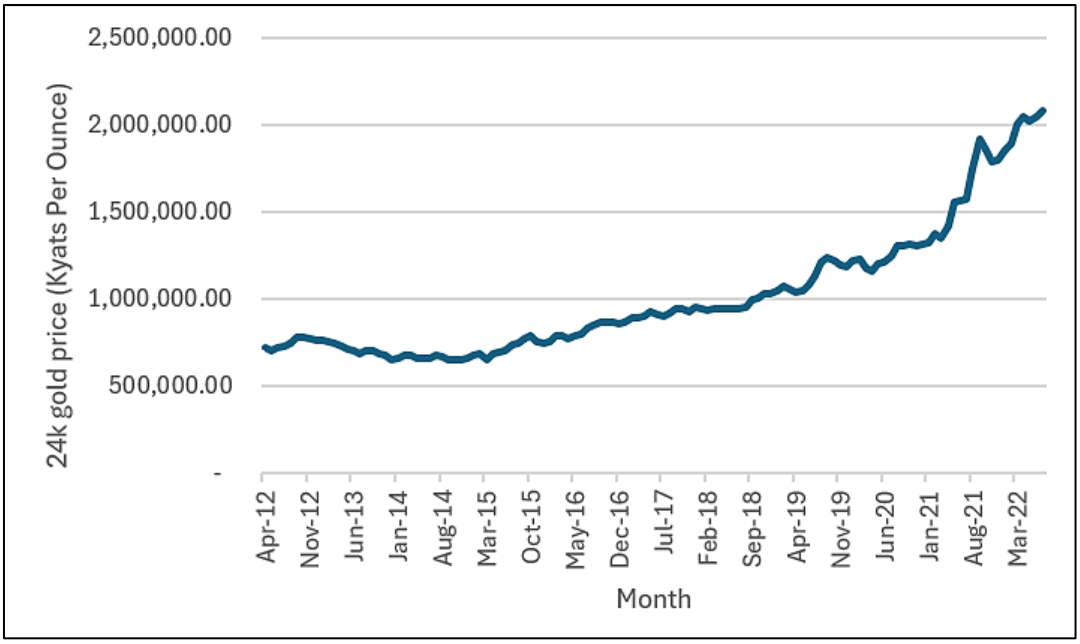
- Unidirectional Granger Causality from variable y_t to variable x_t
- Unidirectional Granger Causality from variable x_t to variable y_t
- Bi-directional Granger Causality
- No Causality.

CHAPTER IV
ANALYSIS OF THE RELATIONSHIP BETWEEN 24 CARAT GOLD PRICE AND CONSUMER PRICE INDEX

This chapter presents the empirical results of the study examining the relationship between 24k gold price and consumer price index (CPI) in Myanmar. The analysis is divided into sections such as descriptive statistics, stationary test, Johansen co-integration test for long-run relationships, estimation of the Vector Autoregression (VAR) Model, Lagrange-Multiplier Test, and Granger Causality test examines causal directions among the variables.

4.1 Descriptive Statistics of 24 Carat Gold Price and Inflation in Myanmar

The study examines monthly data on 24 Carat gold prices and CPI from April 2012 to July 2022, can be found at Appendix.



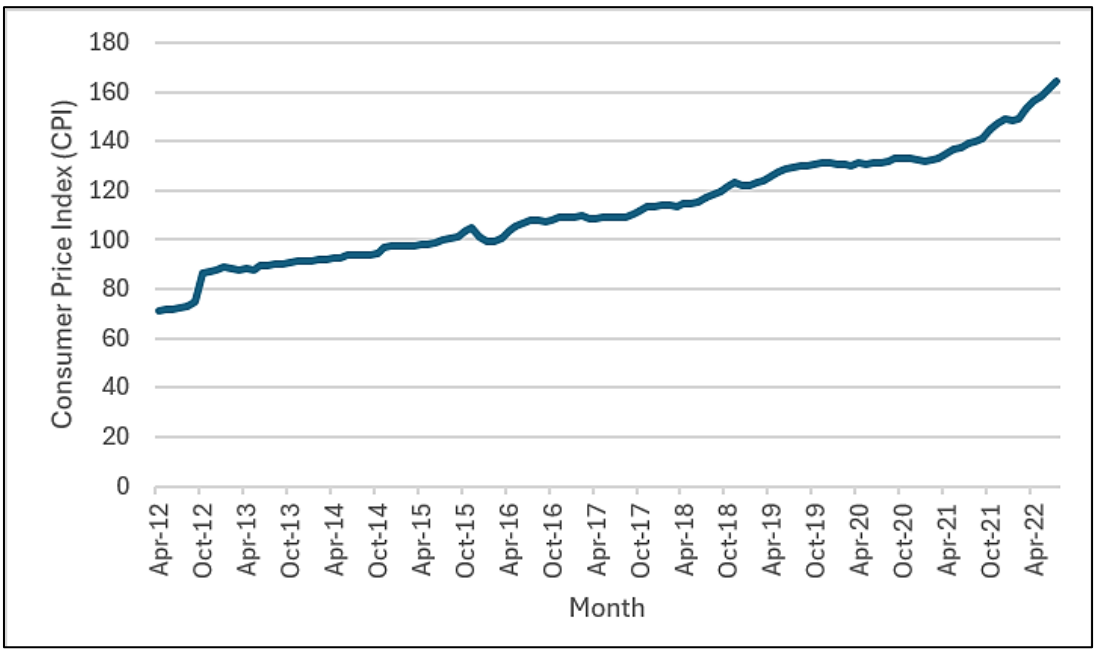
Source: Central Statistical Organization (CSO) Myanmar

Figure (4.1) Gold Price (Kyats per Ounce) in Myanmar

Figure (4.1) illustrates the historical trend of the 24k gold price, denominated in Myanmar Kyats per ounce, spanning from April 2012 to March 2022. A general upward trajectory in the price of gold is observable throughout the depicted period.

Initially, from April 2012 to approximately August 2014, the gold price exhibits some volatility but largely remains within a range of K650,000 to K800,000 per ounce, with a noticeable dip around mid-2014. Following this, from late 2014 to late 2016, there is a gradual and consistent increase in the gold price, breaking past the K800,000 mark and approaching K1,000,000 per ounce.

The period from early 2017 to early 2020 shows a continued, albeit more pronounced, upward trend, with the price surpassing K1,000,000 and reaching approximately K1,250,000 per ounce by early 2020. A significant acceleration in the gold price is evident from early 2020 onwards, with a steep incline particularly in mid-2020 and again in late 2021. By March 2022, the price of 24k gold in Myanmar Kyats per ounce has surged beyond K2,000,000, reaching the highest point in the observed decade. This sharp appreciation in the latter part of the period suggests potential influences from various economic factors, including but not limited to, domestic inflation, global economic uncertainties, and currency fluctuations.



Source: Central Statistical Organization (CSO) Myanmar

Figure (4.2) CPI in Myanmar

Figure (4.2) shows the trend of the Consumer Price Index (CPI) over a decade, from April 2012 to April 2022. The CPI, a key indicator of inflation, shows a consistent upward trajectory throughout this period, signifying a general increase in the price level of goods and services. Initially, from April 2012 to approximately late 2014, the CPI

demonstrates a relatively stable growth, moving from an index value around 70 to just below 100. A more pronounced increase is observed from late 2014 through 2015, where the index surpasses the 100 marks. Subsequently, from early 2016 to late 2018, the CPI continues its ascent, albeit at a slightly slower pace, fluctuating around values between 110 and 125.

A significant acceleration in the CPI's growth becomes apparent from late 2018 onwards, with the index consistently rising and breaking past 130 by early 2020. This upward momentum intensifies particularly from late 2020 through to April 2022, culminating in the CPI reaching approximately 160. The steepness of the curve in the latter part of the period indicates a higher rate of inflation compared to the earlier years. This sustained increase in the CPI suggests a continuous erosion of purchasing power over the decade, with the most significant inflationary pressures observed in the more recent years of the dataset.

Table (4.1) Descriptive Statistics of 24 Carat Gold Price and CPI in Myanmar from April 2012 to July 2022

Description	24 Carat Gold Price (Kyats per Ounce)	CPI (Index)
Mean	1017108.359	112.609
Median	907751	109.2422
Mode	945442	70.9956
Standard Deviation	375288.2361	21.0892
Kurtosis	0.9279	-0.5935
Skewness	1.3237	0.2345
Range	1430605	93.5895
Minimum	649458	70.9956
Maximum	2080063	164.5851
Sum	126121436.5	13963.516
Observations	124	124

Source: Central Statistical Organization (CSO) Myanmar

Table (4.1) shows descriptive statistics of 24 Carat Gold Price and CPI in Myanmar from April 2012 to July 2022 of two key economic indicators: the 24 Carat

Gold Price (in Kyats per Ounce) and the Consumer Price Index (CPI Index) for Myanmar.

For the 24 Carat Gold Price, the mean value is 1,017,108.359 Kyats per ounce, while the median is 907,751 Kyats, indicating that the data is positively skewed, as the mean is higher than the median. The mode is 945,442 Kyats. The standard deviation of 375,288.2361 suggests significant variability in gold prices over the period. The range, calculated as the difference between the maximum (2,080,063 Kyats) and minimum (649,458 Kyats) prices, is 1,430,605 Kyats, confirming a wide fluctuation. The positive skewness of 1.3237 further supports the observation that the distribution of gold prices is skewed to the right, meaning there were more instances of lower prices with a tail extending to higher prices. The kurtosis of 0.9279 indicates that the distribution is platykurtic, meaning it has lighter tails and fewer extreme outliers compared to a normal distribution.

Regarding the CPI (Index), the mean value is 112.609, with a median of 109.2422, again suggesting a slight positive skewness, although less pronounced than for gold prices. The mode is 70.9956. The standard deviation of 21.0892 indicates a moderate level of dispersion around the mean. The range for the CPI is 93.5895, calculated from a minimum of 70.9956 and a maximum of 164.5851. The skewness of 0.2345 confirms a mild positive skew. The negative kurtosis of -0.5935 suggests that the CPI distribution is also platykurtic, similar to gold prices, indicating a relatively flat distribution with fewer extreme values. Both datasets comprise 124 observations, sourced from the Central Statistical Organization (CSO) Myanmar.

4.2 Testing the Stationarity of 24k Gold Price and CPI in Myanmar

Before investigating the causality test among the variables, it's necessary to determine whether the study's variables are stationary or non-stationary. In this study there are two different forms of unit root tests such as Augmented Dicky Fuller test and Phillips Perron test are employed for all the variables.

H_0: The 24k gold price and CPI have unit root. (Non-stationary)

H_1: The 24k gold price and CPI haven't unit root (Stationary)

The results of Augmented Dicky Fuller test and Phillips Perron test at level are presented in Table (4.2) and (4.3).

Table (4.2) Augmented Dickey Fuller and Phillips Perron Tests at Level

Variables	ADF Test		PP Test		Critical Values		
	t-stat	Prob.*	t-stat	Prob.*	1%	5%	10%
Log 24k GP	-1.234	0.654	-1.198	0.671	-4.032	-3.447	-3.147
CPI	-1.876	0.342	-1.912	0.326	-4.032	-3.447	-3.147

Source: Central Statistical Organization (CSO) Myanmar.

Table (4.2) presents the results of the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) unit root tests for the 24k gold price (24k GP) and Consumer Price Index (CPI) at their levels. Both tests are used to determine the stationarity of a time series, where the null hypothesis (H_0) assumes the presence of a unit root (non-stationarity), and the alternative hypothesis (H_1) indicates stationarity.

For the 24k gold price, the ADF test yields a t-statistic of 0.385 with a p-value of 0.997, and the PP test returns a t-statistic of 1.444 with a p-value of 0.999. Similarly, for CPI, the ADF test gives a t-statistic of -0.164 and a p-value of 0.992, while the PP test shows a t-statistic of -0.949 with a p-value of 0.9507. All p-values are substantially higher than the conventional significance levels (1%, 5%, and 10%), and none of the test statistics exceed the corresponding critical values in absolute terms. Therefore, test fails to reject the null hypothesis at level for both variables, indicating that both the 24k gold price and CPI are non-stationary in their original form.

Table (4.3) Augmented Dickey Fuller and Phillips Perron Tests at First Difference

Variables	ADF Test		PP Test		Critical Values		
	t-stat	Prob.*	t-stat	Prob.*	1%	5%	10%
Δ Log 24k GP	-8.921	0.000	-8.740	0.000	-4.033	-3.447	-3.147
Δ CPI	-8.370	0.000	-8.349	0.000	-4.033	-3.447	-3.147

Source: Central Statistical Organization (CSO) Myanmar.

Table (4.3) shows the ADF and PP test results after first differencing the 24k gold price and CPI series. This is done to examine whether the series become stationary after removing trends or other forms of non-stationarity.

For the first-differenced 24k gold price, the ADF test produces a t-statistic of -8.921 with a p-value of 0.000, while the PP test gives a t-statistic of -8.483, also with a p-value of 0.000. For the first-differenced CPI, the ADF and PP test statistics are -8.740

and -8.349 respectively, both with p-values of 0.000. These test statistics are all well below the 1% critical value of -3.487, and the p-values are less than 0.01. Hence, the test strongly rejects the null hypothesis at the first difference, indicating that both the 24k gold price and CPI become stationary after first differencing.

4.3 VAR Lag Length Selection of 24k Gold Price and CPI in Myanmar

At the time series are integrated with the same order, the Vector Auto Regression (VAR) lag order selection method is used to choose the optimal lag length Number of criteria such as Hannan-Quinn Information (HQ), Bayesian Information Criterion (BIC), Akaike Information Criteria (AIC) and Schwarz Information Criteria (SIC) are used to detect the lag length. The lag specification criteria results are shown in Table (4.4).

Table (4.4) Lag Length Selection of 24k Gold Price and CPI in Myanmar

Lag	LL	LR	FPE	AIC	HQIC	SBIC
0	49.240600		0.001570	-0.780837	-0.762068	-0.734625
1	61.294800	24.108000	0.001374	-0.919363	-0.857658	-0.775329*
2	66.583200	10.487*	0.001347*	-0.934516*	-0.840675*	-0.703459

Source: Central Statistical Organization (CSO) Myanmar.

After conducting the unit root or stationarity tests, the optimal lag length for the Vector Autoregression (VAR) model is determined using several lag selection criteria. Table (4.6) presents the results of the VAR lag order selection based on various statistical measures. To determine the optimal lag length for the VAR model, several information criteria were evaluated, including the Final Prediction Error (FPE), Akaike Information Criterion (AIC), Hannan–Quinn Information Criterion (HQIC), and Schwarz Bayesian Information Criterion (SBIC), along with the Likelihood Ratio (LR) test. The results indicate that lag 2 is optimal according to the FPE, AIC, and HQIC. Although SBIC suggests lag 1 as optimal, it is generally more conservative due to its heavier penalty on model complexity. Furthermore, the LR test shows that including lag 2 significantly improves the model fit compared to lag 1 ($p = 0.033$). Based on the majority of the selection criteria and the statistical significance from the LR test, lag 2

was chosen as the appropriate lag length for the VAR model, ensuring a balance between model accuracy and parsimony.

4.4 Johansen Cointegration Test of 24k Gold Price and CPI in Myanmar

Since, the Johansen Cointegration test is applied to determine whether there is cointegration as well as whether there are long run relationship between 24k Gold Price and CPI. The results of Johansen Cointegration test results are presented in Table (4.5).

H_0: There is no long run relationship between 24k Gold Price and CPI.

H_1: There is long run relationship between 24k Gold Price and CPI.

Table (4.5). Johansen Cointegration Test

Cointegration Test		
Hypothesized No. of CE(s)	Trace Statistic	0.05 Critical Value
None	6.214	15.495
At Most 1	1.893	3.841

Source: Central Statistical Organization (CSO) Myanmar.

The Johansen cointegration test results reveal no long-term equilibrium relationship between 24k Gold Price and CPI. For the null hypothesis of "None", the trace statistic (6.214) is below the 5% critical value (15.495), failing to reject H_0 ($p>0.05$). Similarly, for the hypothesis of "At Most 1" cointegrating relationship, the trace statistic (1.893) is also below its critical value (3.841).

4.5 Specification and Estimation of VAR Model

The below Table (4.6) explores whether the relationship between gold price fluctuations and CPI changes to determine if they influence each other.

Table (4.6) VAR Estimation Result

Equation	Variable	Lag	Coefficient	Std. Err.	z-value	p-value
Δ Log Gold	Δ Log Gold	L1	0.3282293***	0.0874847	3.75	0.000
	Δ Log Gold	L2	-0.2662307***	0.0897776	-2.97	0.003
	Δ CPI	L1	0.0002804	0.0016206	0.17	0.863
	Δ CPI	L2	0.000475	0.0015807	0.3	0.764
	cons	-	0.0076047***	0.0027574	2.76	0.006
Δ CPI	Δ Log Gold	L1	12.18184**	4.921236	2.48	0.013
	Δ Log gold	L2	6.640249	5.050219	1.31	0.189
	ΔCPI	L1	0.2048283**	0.0911644	2.25	0.025
	Δ CPI	L2	0.0250183	0.0889184	0.28	0.778
	cons	-	0.4318745***	0.1551099	2.78	0.005

Source: Central Statistical Organization (CSO) Myanmar.

According to the Table (4.7), VAR equation for 24k Gold is

$$\Delta \text{ Log Gold}_t = 0.0076 + 0.3282\,\Delta \text{ Log Gold}_{t-1} - 0.2662\Delta \text{ Log Gold}_{t-2} + 0.0003\Delta CPI_{t-1} + 0.0005\Delta CPI_{t-2} + \epsilon_{1,t}$$

According to the Table (4.7), VAR equation for CPI is

$$\Delta CPI_t = 0.4319 + 12.1818\Delta \text{ Log Gold}_{t-1} + 6.6402\Delta \text{ Log Gold}_{t-2} + 0.2048\Delta CPI_{t-1} + 0.0250\Delta CPI_{t-2} + \epsilon_{2,t}$$

The VAR model reveals distinct dynamics between gold price changes (Δ Log Gold) and inflation changes (ΔCPI). The findings suggest that Δ Log Gold exhibits a self-correcting pattern, with its first lag (L1) exerting a positive and statistically significant effect (coefficient = 0.328, $p < 0.001$), while the second lag (L2) has a negative and significant impact (coefficient = -0.266, $p = 0.003$). This implies short-term persistence followed by a reversal in gold price movements, consistent with mean-reverting behavior in commodity markets. Notably, Δ CPI lags do not significantly influence Δ Log Gold (coefficients ≈ 0, $p > 0.8$), indicating that inflation fluctuations do not drive short-term gold price changes.

Conversely, Δ CPI responds strongly to past changes in gold prices. The first lag of Δ Log Gold (L1) has a large and statistically significant positive effect (coefficient = 12.182, $p = 0.013$), suggesting that rising gold prices precede inflationary pressures. This aligns with the theoretical view that gold, as an inflation hedge, may signal future

inflation trends. However, the second lag (L2) of Δ Log Gold is insignificant (p = 0.189), implying that the inflationary impact of gold prices is short-lived. Additionally, Δ CPI displays weak persistence, with only its first lag (L1) showing marginal significance (coefficient = 0.205, p = 0.025), while the second lag (L2) is negligible (p = 0.778).

The intercept terms for both equations are statistically significant (Δ Log Gold: 0.008, p = 0.006; Δ CPI: 0.432, p = 0.005), confirming baseline drift in both series. Overall, the results suggest unidirectional causality from gold prices to CPI or inflation, supporting the hypothesis that gold serves as a leading indicator for inflationary trends, whereas inflation does not significantly feed back into gold price dynamics in the short run.

4.6 Stability Analysis of VAR Model Eigenvalues

The below Table (4.7) shows the Eigenvalue Stability to confirm the VAR model satisfies the stability condition.

Table (4.7) Eigenvalue Stability Check

Eigenvalue	Modulus
0.1472717 + 0.49930291i	0.520569
0.1472717 - 0.49930291i	0.520569
0.3430464	0.343046
-0.1053312	0.105331

Source: Central Statistical Organization (CSO) Myanmar.

The Eigenvalue Stability Check confirms that the estimated VAR model satisfies the stability condition, as all eigenvalues lie strictly inside the unit circle. The results show four eigenvalues with moduli ranging between 0.105331 and 0.520569, all significantly below the critical threshold of 1. This indicates that the VAR system is covariance stationary and will produce impulse response functions that decay over time rather than explode.

The presence of complex conjugate eigenvalues (0.1472717 ± 0.49930291i) suggests oscillatory but ultimately stable dynamics in the system, with the modulus of 0.520569 governing the rate of decay. The purely real eigenvalues (0.343046 and -0.105331) represent direct autoregressive effects, where the negative eigenvalue

implies a slight alternating pattern in the adjustment process. The fact that the largest modulus is 0.520569 (significantly <1) provides strong evidence that the VAR specification is appropriate for forecasting and structural analysis, as shocks to the system will gradually dissipate rather than accumulate.

4.7 Summary Statistics and Diagnostics of Residuals

The following Table (4.8) and subsequent analysis provide a comprehensive overview of the model's residuals, confirming its robustness and adherence to critical statistical assumptions.

Table (4.8) Error Distribution and Diagnostic Summary

Variable	Mean	Std. Dev.	Min	Max
error	3.73e-11	0.024615	-0.053156	0.092033

Source: Central Statistical Organization (CSO) Myanmar.

The descriptive statistics of the model residuals indicate that the VAR specification is well-behaved and satisfies key diagnostic assumptions. The mean residual is effectively zero, suggesting the model produces unbiased estimates without systematic over- or under-prediction. The standard deviation of 0.0246 reflects moderate dispersion in the forecast errors, while the range between -0.0532 and 0.0920 shows a roughly symmetric distribution of residuals around the mean.

The near-zero mean combined with the relatively small standard deviation indicates proper model specification with no major omitted variables or structural breaks. The symmetry of the minimum and maximum values suggests the residuals are approximately normally distributed, an important condition for valid statistical inference. The time series plot of residuals oscillating evenly around the zero line further confirms the absence of heteroskedasticity or autocorrelation patterns. These residual diagnostics collectively support the reliability of the VAR model for both explanatory analysis and forecasting purposes.

4.8 Lagrange-Multiplier Test for Autocorrelation

To ensure the reliability of the VAR Model, it is important to examine whether the residuals are free from serial correlation. The Lagrange Multiplier (LM) test is employed to detect autocorrelation in the residuals up to a specified lag order. This

diagnostic test helps validate the model's specification by checking if the residuals resemble white noise.

The hypothesis is:

H_0: There is no serial correlation.

H_1: There is serial correlation

Table (4.9) Lagrange-Multiplier Test

Lag	χ^2 Statistic	Degrees of Freedom (df)	p-value
1	5.9627	4	0.20195
2	5.6093	4	0.23029

Source: Central Statistical Organization (CSO) Myanmar.

The Lagrange-multiplier (LM) test results indicate no evidence of autocorrelation in the VAR model residuals at lag orders 1 and 2. For lag 1, the χ^2 statistic of 5.9627 (df = 4) yields a p-value of 0.202, and for lag 2, the χ^2 statistic of 5.6093 (df = 4) gives a p-value of 0.230. Both p-values exceed conventional significance levels of 1 percent, 5 percent, and 10 percent, indicating that the null hypothesis cannot be rejected at any lag. This suggests that the VAR model is well-specified and uncorrelated across all tested lag lengths, thereby supporting the model's adequacy for inference and forecasting.

4.9 Causality Between 24k Gold Price and CPI in Myanmar

The Granger Causality test is employed to identify the causal relationship between the two variables of 24k Gold Price and CPI in Myanmar. As the p value of one variable, CPI was significantly to forecast the value of the variable of 24k gold price, then it can be stated that the CPI granger cause with the gold price. The results of Pairwise Granger Causality test are shown in the Table (4.10).

H_0: There is no causal relationship between 24k Gold Price and CPI.

H_1: There is a causal relationship between 24k Gold Price and CPI.

Table (4.10) Results of Causality Between 24k Gold Price and CPI

Particular	Observation	F-statistics	Prob.**
24k Gold Price does not Granger cause CPI	124	10.091	0.006
CPI does not Granger cause 24k Gold Price	124	0.15208	0.927

Source: Central Statistical Organization (CSO) Myanmar.

The Granger Causality test is conducted to determine the directional causal relationship between the 24k Gold Price (Δ Log Gold) and the CPI (Δ CPI) in Myanmar. The null hypothesis (H_0) states that there is no causal relationship between the variables, while the alternative hypothesis (H_1) suggests the presence of a causal relationship.

The results of the Granger causality test reveal a unidirectional causal relationship originating from Δ Log Gold to Δ CPI. Specifically, changes in 24k Gold Price are found to Granger-cause changes in the Consumer Price Index (χ^2=10.091, p-value =0.006), implying that historical gold price movements hold significant predictive power for future inflation dynamics in Myanmar. Conversely, changes in the Consumer Price Index do not Granger-cause changes in 24k Gold Price (χ^2=0.152, p-value =0.927). This asymmetry suggests that 24k Gold Price may serve as a leading indicator for inflation in the Myanmar context, while inflationary shocks do not immediately feed back into the gold market. These findings generally align with the perception of gold as an asset that can reflect or anticipate inflationary pressures in financial markets.

CHAPTER V
CONCLUSION

This chapter describes the key findings and provides discussion of the results obtained from the study. It also outlines suggestions and recommendations based on the findings. In addition, this chapter offers guidance for future researchers who wish to explore similar topics or build upon this study.

5.1 Findings

This study analyzed the dynamic relationship between 24 Carat gold prices and the Consumer Price Index (CPI), in Myanmar over the period from April 2012 to July 2022. The investigation covered various econometric approaches, including descriptive statistics, stationarity tests, Vector Auto Regression (VAR), cointegration analysis, and causality testing.

The descriptive statistics revealed that both gold prices and CPI exhibited an upward trend over the study period. Gold prices showed significant volatility, with a mean value of 1,017,108 Kyats per ounce and a wide range between 649,458 and 2,080,063 Kyats. The positive skewness indicated that extreme price increases were more common than decreases. Similarly, CPI displayed a consistent rise, reflecting inflationary pressures in Myanmar's economy. The mean CPI was 112.609, with a gradual but accelerating increase, particularly after 2020, suggesting worsening inflation.

Unit root tests (ADF and PP) confirmed that both gold prices and CPI were non-stationary at levels but became stationary after first differencing. This finding supported the use of a Vector Autoregression (VAR) model to analyze their short-term interactions. The optimal lag length for the VAR model was determined to be two, based on information criteria such as AIC and HQIC. The Johansen cointegration test indicated no long-run equilibrium relationship between 24 K gold price and CPI, meaning they do not move together in the long term. The VAR model revealed short-term dynamics. The stability of the VAR model was confirmed through eigenvalue analysis, as all eigenvalues lay within the unit circle, ensuring that the model was stable for forecasting. Residual diagnostics showed no autocorrelation, further validating the model's reliability.

The Granger causality test provided crucial evidence of a unidirectional relationship: gold prices Granger-caused CPI, but not vice versa. This implies that gold prices act as a leading indicator for inflation in Myanmar, whereas inflation does not directly influence gold price movements in the short term. This finding aligns with the theory that gold serves as a hedge against inflation, as investors may turn to gold during periods of expected price increases, driving up demand and prices.

In summary, the study found that while gold prices and inflation do not share a long-term relationship, short-term fluctuations in gold prices can signal future inflationary trends. Policymakers and investors should monitor gold price movements as an early indicator of potential inflation, while recognizing that inflation itself does not immediately feed back into gold price adjustments. These findings contribute to a better understanding of Myanmar's inflationary dynamics and the role of gold in the economy.

5.2 Suggestions and Recommendations

This study investigated the dynamic relationship between 24 Carat gold prices and the Consumer Price Index (CPI) in Myanmar from April 2012 to July 2022, employing a comprehensive set of econometric approaches. Crucially, the Granger causality test established a unidirectional relationship: gold prices Granger-caused CPI, but not vice versa. This implies that gold prices serve as a leading indicator for inflation in Myanmar, without inflation directly influencing gold price movements in the short term. These findings collectively suggest that while gold prices and inflation do not share a long-term equilibrium, short-term fluctuations in gold prices can provide valuable signals for future inflationary trends in Myanmar's economy.

For policymakers, particularly the Central Bank of Myanmar and fiscal authorities, this suggests a pressing need to monitor gold price trends as part of their macroeconomic surveillance. Incorporating gold price movements into inflation forecasting models could enhance the accuracy and timeliness of monetary policy responses. Since rising gold prices may precede increases in consumer prices, early detection of such trends can support preemptive measures to maintain price stability. Additionally, regulating speculative behavior in the gold market and promoting transparency in pricing mechanisms could help reduce volatility, thereby minimizing gold-induced inflationary shocks.

From an investment perspective, the findings reaffirm gold's traditional role as a hedge against inflation in Myanmar. The lack of short-run causality from inflation to gold prices indicates that gold retains its value independently of immediate inflation fluctuations, making it a reliable store of value in times of economic uncertainty. Investors and financial institutions may consider gold as part of a diversified portfolio to mitigate inflation risks. Furthermore, financial regulators could explore the development of gold-backed financial products, such as gold savings accounts or gold-linked bonds, to provide safe and accessible investment options to the public, especially during periods of macroeconomic instability.

In summary, this study underscores the significance of gold price movements as an early warning signal of inflation in Myanmar. Integrating gold market analysis into economic planning and policy formulation can strengthen the country's capacity to respond effectively to inflationary pressures and promote more informed investment behavior.

5.3 Further Study

While this study provides meaningful insights into the relationship between 24k gold prices and consumer price inflation (CPI) in Myanmar, it is subject to certain limitations. The analysis is confined to monthly data over a specific 10-year period and considers only two macroeconomic variables. Although this bivariate framework offers a focused view of the gold-inflation dynamic, it may overlook broader economic influences that could contribute to or mediate the observed relationship.

To build upon the findings of this study, future research could adopt a multivariate approach by incorporating additional macroeconomic variables such as exchange rates, interest rates, global gold prices, money supply, and geopolitical risks. Including these variables would allow for a more comprehensive understanding of the complex factors influencing inflation and gold price behavior in Myanmar's economy. Moreover, future studies could employ more advanced econometric techniques to capture potential nonlinearities and structural changes in the relationship. For instance, Vector Error Correction Models (VECM) could be applied if cointegration among multiple variables is established, allowing researchers to distinguish between short-run dynamics and long-run equilibrium relationships. Markov Switching VAR models or Threshold VAR models could help detect regime shifts or asymmetric effects in different inflationary environments. Structural VAR (SVAR) methods may also be used

to identify the sources of macroeconomic shocks and distinguish between exogenous and endogenous drivers of inflation and gold price movements.

Additionally, extending the dataset to include post-pandemic periods or major political and economic transition phases could help assess the robustness and evolution of the gold-inflation relationship over time. Panel data approaches comparing Myanmar with other similar economies could also yield valuable cross-country insights into how gold functions as an inflation hedge under varying institutional and market conditions. By addressing these areas, future research can contribute to a more nuanced and policy-relevant understanding of commodity price dynamics and inflation in developing economies.

REFERENCES

Asian Development Bank (2020). Key indicators for Asia and the Pacific 2020. Asian Development Bank.

Asian Development Bank (2021). Asian development outlook 2021: Financing a green and inclusive recovery. Asian Development Bank.

Asian Development Bank (2022). Asian development outlook 2022: Mobilizing taxes for development. Asian Development Bank.

Aung, M. M. (2023). Analyzing the cointegration and causal relationship between export and import in ASEAN countries (1970–2021). United Nations Database.

Aung, N., & Khaing, K. S. (2021). Gold market resilience in Myanmar during financial crises. Myanmar Economic Review, 3(2), 45–58.

Baur, D. G., & Lucey, B. M. (2010). Is gold a hedge or a safe haven? An analysis of stocks, bonds and gold. Financial Review, 45(2), 217–229.

Baur, D. G., & McDermott, T. K. (2010). Is gold a safe haven? International evidence. *Journal of Banking & Finance*, 34(8), 1886–1898.

Baur, D. G., Lucey, B. M., & Pöttmann, G. (2014). Gold and inflation: Hedge or hedge not? Finance Research Letters, 11(2), 86–93.

Beckmann, J., & Czudaj, R. (2013). Gold as an inflation hedge in a time-varying coefficient framework. *The North American Journal of Economics and Finance*, 24, 208–222.

Bekaert, G., & Harvey, C. R. (2017). Emerging equity markets in a globalizing world. *Handbook of the Economics of Finance*, 2, 621–687.

Capie, F., Mills, T. C., & Wood, G. (2005). Gold as a hedge against the dollar. *Journal of International Financial Markets, Institutions and Money*, 15(4), 343–352.

Central Bank of Myanmar (2022). Annual report.

Cheema, M. A., Faff, R., & Szulczuk, K. (2020). The 2008 global financial crisis and COVID-19 pandemic: How safe is safe-haven investing in gold? Finance Research Letters, 38, 101431.

Chua, J., & Woodward, R. S. (1982). Gold as an inflation hedge: A comparative study of six major industrial countries. *Journal of Business Finance & Accounting*, 9(2), 191–200.

Central Statistical Organization. (2022). Selected monthly economic indicators.

Dempster, N. (2006). The role of gold in global asset allocation. World Gold Council.

Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366a), 427–431.

Dolado, J., Jenkinson, T., & Sosvilla-Rivero, S. (1990). Cointegration and unit roots. *Journal of Economic Surveys*, 4(3), 249–273.

Dimitriou, D., & Simos, T. E. (2013). Causal relationship between gold prices and inflation: Evidence from the USA and the Eurozone. Applied Economics Letters, 20(9), 813–817.

Enders, W. (2014). Applied econometric time series (4th ed.). Wiley.

Ghosh, D., Levin, E. J., Macmillan, P., & Wright, R. E. (2004). Gold as an inflation hedge. Studies in Economics and Finance, 22(1), 1–25.

GIZ. (2022). Formalizing Myanmar's gold market: policy recommendations. Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH.

Gujarati, D. N., & Porter, D. C. (2009). Basic econometrics (5th ed.). McGraw-Hill.

Hamilton, J. D. (1994). Time series analysis. Princeton University Press.

International Monetary Fund. (2021). Myanmar country report.

Investopedia. (2023). Consumer price index (CPI): Definition, formula, and uses. Investopedia.

Jain, R. (2012). Gold and inflation: A study of Indian markets. *Indian Journal of Economics and Business,* 11(1), 59–73.

Japan External Trade Organization. (2013). Myanmar market trend report: Retail and consumer goods.

MacKinnon, J. G. (1991). Critical values for cointegration tests. Long-run Economic Relationships: Readings in Cointegration, 267–276. Oxford University Press.

Mahmoud, E. (1984). Accuracy in forecasting: *A survey. Journal of Forecasting*, 3(2), 139–159.

McNees, S. K. (1986). Forecasting accuracy of alternative techniques: A comparison of US macroeconomic forecasts. *Journal of Business & Economic Statistics*, 4(1), 5–15.

Musonera, E. (2020). Time series analysis of gold prices and inflation in Africa. *African Journal of Economic Policy*, 27(2), 56–70.

Narayan, P. K., Narayan, S., & Zheng, X. (2010). Gold and oil futures markets: Are markets efficient? Applied Energy, 87(10), 3299–3303.

Nguyen, Q., & Tran, L. (2020). Inflation and gold price nexus in Vietnam: An empirical investigation. Asian Economic and Financial Review, 10(3), 367–382.

Organisation for Economic Co-operation and Development. (2022). Informality and taxation in Myanmar: Challenges and reform opportunities.

Phillips, P. C. B., & Perron, P. (1988). Testing for a unit root in time series regression. Biometrika, 75(2), 335–346.

Reuters. (2024). Central banks increasing gold holdings amid inflation fears. Reuters News Service.

Salisu, A. A., Akanni, L. O., & Raheem, I. D. (2020). Gold as a hedge and safe haven for uncertainty: A meta-analysis. Resources Policy, 66, 101605.

Sims, C. A. (1980). Macroeconomics and reality. Econometrica, 48(1), 1–48.

Sujit, K. S., & Kumar, B. R. (2011). Study on dynamic relationship among gold price, oil price, exchange rate, and stock market returns. *International Journal of Applied Business and Economic Research*, 9(2), 145–165.

Tiwari, A. K., Shahbaz, M., & Islam, F. (2015). Does gold act as a hedge against inflation in India? A wavelet-based approach. Economic Modelling, 51, 225–233.

Turnell, S. (2011). Fiery dragons: Banks, moneylenders and microfinance in Burma. NIAS Press.

United Nations Development Programme. (2022). Gold and financial inclusion: Pathways to stability in Myanmar.

United Nations Economic and Social Commission for Asia and the Pacific. (2021). Myanmar economic situation report.

Wang, Y., & Lee, Y. W. (2011). Gold as a hedge against inflation: Evidence from global markets. *International Journal of Economics and Finance*, 3(4), 190–198.

Wang, Y., Wu, C., & Yang, L. (2011). Oil price shocks and stock market activities: Evidence from oil-importing and oil-exporting countries. *Journal of Comparative Economics,* 39(3), 475–493.

World Bank (2023). Myanmar economic monitor: Staying afloat. World Bank Group.

World Bank (2024). Myanmar economic monitor: Inflation and exchange rate outlook. World Bank Group.

World Gold Council (2019). Gold demand trends Q4 2019.

World Gold Council (2020). Gold: The most effective commodity investment.

World Gold Council (2021). Retail gold insights: ASEAN market perspective.

World Gold Council (2024). Central bank gold reserves survey.

Zin, N. (2022). Cointegration and causal relationship between gold price and exchange rate in Myanmar (2011–2020) (Master's thesis). Yangon University of Economics.

APPENDICES

Table (A.1) Monthly average 24k Gold price in Myanmar

Year/ Month	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Jan	-	764,731	658,135	680,033	758,740	864,180	944,136	1,050,948	1,217,833	1,324,114	1,858,600
Feb	-	752,783	676,352	684,050	787,708	890,563	940,922	1,070,978	1,227,896	1,374,206	1,893,950
Mar	-	743,705	680,552	650,778	792,300	892,528	945,042	1,059,125	1,178,043	1,347,500	2,006,318
Apr	719,667	726,472	664,271	683,944	774,382	905,452	942,633	1,042,705	1,163,300	1,423,261	2,050,188
May	702,108	709,284	661,084	697,500	787,608	928,050	945,026	1,047,588	1,204,188	1,557,523	2,022,782
Jun	718,154	702,908	660,820	705,260	798,846	910,050	947,050	1,084,932	1,216,685	1,563,260	2,050,691
Jul	731,720	690,172	676,780	741,250	834,860	901,064	950,133	1,132,832	1,248,700	1,576,550	2,080,063
Aug	744,346	702,641	671,665	747,320	851,848	916,041	955,892	1,215,558	1,309,300	1,746,453	-
Sep	780,988	705,580	657,115	769,548	870,980	945,442	994,244	1,242,720	1,310,135	1,916,357	-
Oct	778,935	688,135	655,080	789,308	868,135	945,442	1,009,813	1,221,091	1,311,958	1,858,571	-
Nov	770,271	675,592	649,458	756,309	870,438	931,648	1,034,817	1,191,326	1,308,605	1,788,763	-
Dec	764,392	652,560	664,038	744,638	863,704	950,400	1,031,978	1,184,152	1,313,375	1,795,786	-
Min	**702,108**	**652,560**	**649,458**	**650,778**	**758,740**	**864,180**	**940,922**	**1,042,705**	**1,163,300**	**1,324,114**	**1,858,600**
Max	**780,988**	**764,731**	**680,552**	**789,308**	**870,980**	**950,400**	**1,034,817**	**1,242,720**	**1,313,375**	**1,916,357**	**2,080,063**

Table (A.2) Monthly CPI in Myanmar

Year/ Month	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Jan	-	89	92	98	100	109	114	122	131	132	148
Feb	-	88	92	98	100	110	114	123	130	132	149
Mar	-	88	92	98	100	108	114	124	130	133	153
Apr	71	88	92	98	104	108	114	126	131	135	156
May	72	88	93	98	105	109	115	127	131	137	158
Jun	72	90	94	99	107	109	115	129	131	138	161
Jul	72	90	94	100	108	109	117	130	131	139	165
Aug	73	90	94	100	108	109	118	130	132	140	-
Sep	75	90	94	101	108	110	120	130	133	141	-
Oct	87	91	94	104	108	112	121	131	133	145	-
Nov	87	91	97	105	109	113	123	131	133	147	-
Dec	88	92	97	101	109	114	122	131	132	149	-
Min	**71**	**88**	**92**	**98**	**100**	**108**	**114**	**122**	**130**	**132**	**148**
Max	**88**	**92**	**97**	**105**	**109**	**114**	**123**	**131**	**133**	**149**	**165**

Table (A.3) Stationary test

Dickey-Fuller test for unit root Number of obs = 122

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-8.921	-4.033	-3.447	-3.147

MacKinnon approximate p-value for Z(t) = 0.0000

D.diff_lgold	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
diff_lgold						
L1.	-.7987696	.0895347	-8.92	0.000	-.9760572	-.621482
_trend	.0001449	.0000672	2.16	0.033	.0000119	.000278
_cons	-.0017391	.0046067	-0.38	0.706	-.0108609	.0073827

Phillips-Perron test for unit root Number of obs = 122
Newey-West lags = 4

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(rho)	-82.280	-27.547	-20.788	-17.573
Z(t)	-8.740	-4.033	-3.447	-3.147

MacKinnon approximate p-value for Z(t) = 0.0000

diff_lgold	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
diff_lgold						
L1.	.2012304	.0895347	2.25	0.026	.0239428	.378518
_trend	.0001449	.0000672	2.16	0.033	.0000119	.000278
_cons	-.0017391	.0046067	-0.38	0.706	-.0108609	.0073827

Dickey-Fuller test for unit root Number of obs = 122

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-8.370	-4.033	-3.447	-3.147

MacKinnon approximate p-value for Z(t) = 0.0000

D.diff_cpi	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
diff_cpi						
L1.	-.7528479	.0899408	-8.37	0.000	-.9309396	-.5747563
_trend	.0034329	.0037205	0.92	0.358	-.003934	.0107999
_cons	.3679044	.2674252	1.38	0.171	-.1616241	.8974329

Phillips-Perron test for unit root — Number of obs = 122
Newey-West lags = 4

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(rho)	-90.828	-27.547	-20.788	-17.573
Z(t)	-8.349	-4.033	-3.447	-3.147

MacKinnon approximate p-value for Z(t) = 0.0000

diff_cpi	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
diff_cpi						
L1.	.2471521	.0899408	2.75	0.007	.0690604	.4252437
_trend	.0034329	.0037205	0.92	0.358	-.003934	.0107999
_cons	.3679044	.2674251	1.38	0.171	-.1616241	.8974329

Table (A.4) VAR Lag length Selection

Selection-order criteria
Sample: 2012m7 - 2022m7 — Number of obs = 121

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	49.2406				.00157	-.780837	-.762068	-.734625
1	61.2948	24.108	4	0.000	.001374	-.913963	-.857658*	-.775329*
2	66.5382	10.487*	4	0.033	.001347*	-.934516*	-.840675	-.703459

Endogenous: diff_lgold diff_cpi
Exogenous: _cons

Table (A.5) VAR Models Estimation

Vector autoregression

Sample: 2012m7 - 2022m7		No. of obs	= 121
Log likelihood =	66.53824	AIC	= -.9345163
FPE =	.0013466	HQIC	= -.8406752
Det(Sigma_ml) =	.0011413	SBIC	= -.7034592

Equation	Parms	RMSE	R-sq	chi2	P>chi2
diff_lgold	5	.025035	0.1335	18.64258	0.0009
diff_cpi	5	1.4083	0.1348	18.85284	0.0008

	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]
diff_lgold						
diff_lgold						
L1.	.3282293	.0874847	3.75	0.000	.1567625	.4996962
L2.	-.2662307	.0897776	-2.97	0.003	-.4421915	-.0902698
diff_cpi						
L1.	.0002804	.0016206	0.17	0.863	-.002896	.0034567
L2.	.000475	.0015807	0.30	0.764	-.0026231	.0035731
_cons	.0076047	.0027574	2.76	0.006	.0022003	.0130091
diff_cpi						
diff_lgold						
L1.	12.18184	4.921236	2.48	0.013	2.536396	21.82729
L2.	6.640249	5.050219	1.31	0.189	-3.257997	16.5385
diff_cpi						
L1.	.2048283	.0911644	2.25	0.025	.0261494	.3835073
L2.	.0250183	.0889184	0.28	0.778	-.1492585	.1992951
_cons	.4318745	.1551099	2.78	0.005	.1278647	.7358843

Eigenvalue stability condition

Eigenvalue	Modulus
.1472717 + .4993029*i*	.520569
.1472717 - .4993029*i*	.520569
.3438454	.343845
-.1053312	.105331

All the eigenvalues lie inside the unit circle.
VAR satisfies stability condition.

Variable	Obs	Mean	Std. Dev.	Min	Max
error	121	3.73e-11	.0246145	-.0531564	.092033

```
. tsline error, yline( 3.73e-11 )
```

Lagrange-multiplier test

lag	chi2	df	Prob > chi2
1	5.9627	4	0.20195
2	5.6093	4	0.23029

H0: no autocorrelation at lag order

Table (A.5) Granger causality test

Granger causality Wald tests

Equation	Excluded	chi2	df	Prob > chi2
diff_lgold	diff_cpi	.15208	2	0.927
diff_lgold	ALL	.15208	2	0.927
diff_cpi	diff_lgold	10.091	2	0.006
diff_cpi	ALL	10.091	2	0.006